Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $\mathrm{C} = 2\cos\theta$, then the value of the determinant $\Delta = \begin{bmatrix} \mathrm{C} & 1 & 0 \\ 1 & \mathrm{C} & 1 \\ 6 & 1 & \mathrm{C} \end{bmatrix}$ is

(A) $\frac{2\sin^2 2\theta}{\sin\theta}$

(B) $8\cos^3\theta - 4\cos\theta + 6$

(C) $\frac{2\sin 2\theta}{\sin\theta}$

(D) $8\cos^3\theta + 4\cos\theta + 6$

Q.2

If $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$, then the value of $\begin{vmatrix} 1 & \mathrm{e}^{\mathrm{i}\pi/3} & \mathrm{e}^{\mathrm{i}\pi/4} \\ \mathrm{e}^{-\mathrm{i}\pi/3} & 1 & \mathrm{e}^{\mathrm{i}2\pi/3} \\ \mathrm{e}^{-\mathrm{i}\pi/4} & \mathrm{e}^{-\mathrm{i}2\pi/3} & 1 \end{vmatrix}$ is

(A) $-2 + \sqrt{2}$

(B) $2 - \sqrt{2}$

(C) $-2 - \sqrt{2}$

(D) 1

Q.3

If the value of the determinant $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix}$ is positive, where $a \neq b \neq c$, then the value of abc

(A) cannot be less than 1

(B) is greater than -8

(C) is less than -8

(D) must be greater than 8

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Q.4

Let $\Delta = \begin{vmatrix} 1+x_1y_1 & 1+x_1y_2 & 1+x_1y_3 \\ 1+x_2y_1 & 1+x_2y_2 & 1+x_2y_3 \\ 1+x_3y_1 & 1+x_3y_2 & 1+x_3y_3 \end{vmatrix}$, then value of Δ is

(A) $x_1x_2x_3 + y_1y_2y_3$

(B) $x_1x_2x_3y_1y_2y_3$

(C) $x_2x_3y_2y_3 + x_3x_1y_3y_1 + x_1x_2y_1y_2$

(D) 0

Q.5

If $a^2+b^2+c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ then $f(x)$ is a polynomial of degree

(A) 1

(B) 0

(C) 3

(D) 2

Q.6

If $\begin{vmatrix} x^2+x & 3x-1 & -x+3 \\ 2x+1 & 2+x^2 & x^3-3 \\ x-3 & x^2+4 & 3x \end{vmatrix}$ $= a_0 + a_1x + a_2x^2 + \ldots + a_7x^7$, then the value of a_0 is

(A) 25

(B) 24

(C) 23

(D) 21

Q.7

Let $x < 1$, then value of $\begin{vmatrix} x^2+2 & 2x+1 & 1 \\ 2x+1 & x+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$ is

(A) Non-negative

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(B) Non-positive

(C) Negative

(D) Positive

Q.8

If [ ] denotes the greatest integer less than or equal to the real number under consideration and $-1 \leq x < 0; 0 \leq y < 1; 1 \leq z < 2$, then the value of the determinant $\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$ is

(A) [z]

(B) [y]

(C) [x]

(D) None of these

Q.9

$A^{-1} = \begin{bmatrix} 1 & -2 \\ -2 & 2 \end{bmatrix}$, what is det (A)?

(A) 2

(B) -2

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

Q.10

If $\mathrm{A} = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$, then $\det(\mathrm{adj}(\mathrm{adj}\ \mathrm{A}))$ is

(A) $(14)^4$

(B) $(14)^3$

(C) $(14)^2$

(D) $(14)^1$

Q.11

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If $\mathrm{f(x)=a+bx+cx^2}$ and α, β, λ are roots of the equation $x^3=1$, then

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

is equal to

(A) $\mathrm{f}(\alpha)+\mathrm{f}(\beta)+\mathrm{f}(\lambda)$

(B) $f(\alpha)f(\beta)+f(\beta)f(\lambda)+f(\gamma)+f(\alpha)$

(C) $\mathrm{f}(\alpha)\mathrm{f}(\beta)\mathrm{f}(\gamma)$

(D) $-f(\alpha)f(\beta)f(\gamma)$

Q.12

If ω is the cube root of unity, then what is one root of the equation

$$\begin{vmatrix} x^2 & -2x & -2\omega^2 \\ 2 & \omega & -\omega \\ 0 & \omega & 1 \end{vmatrix} = 0?$$

(A) 1

(B) -2

(C) 2

(D) ω

Q.13

The value of the determinant

$$\begin{vmatrix} \cos^2 54^\circ & \cos^2 36^\circ & \cot 135^\circ \\ \sin^2 53^\circ & \cot 135^\circ & \sin^2 37^\circ \\ \cot 135^\circ & \cos^2 25^\circ & \cos^2 65^\circ \end{vmatrix}$$ is equal to

(A) -2

(B) -1

(C) 0

(D) 1

Q.14

If $\begin{vmatrix} x^n & x^{n+2} & x^{2n} \\ 1 & x^a & a \\ x^{n+5} & x^{a+6} & x^{2n+5} \end{vmatrix} = 0 \forall x \in R$, where $n \in N$ then value of ' a ' is

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(A) n

(B) $\mathrm{n}-1$

(C) $\mathrm{n}+1$

(D) None of these

Q.15

The value of the determinant

$$\begin{matrix} 1 & a & a^2 \\ \cos(n\text{-}1)x & \cos nx & \cos(n+1)x \\ \sin(n\text{-}1)x & \sin nx & \sin(n+1)x \end{matrix}$$ is zero, if

(A) $\sin x = 0$

(B) $\cos x = 0$

(C) $a = 0$

(D) $\cos x = \frac{1+a^2}{2a}$

Q.16

Let A be the set of all determinants of order 3 with entries 0 or 1 only, B be the subset 0 A consisting of all determinants with value 1 , and C be the subset consisting of all determinants with value $-1'$. Then if $n(B)$ and $n(C)$ denote the number of elements in B and C, respectively, we have

(A) $\mathrm{C} = \phi$

(B) $n(B) = n(C)$

(C) $\mathrm{A} = \mathrm{B} \cup \mathrm{C}$

(D) $n(B) = 2n(C)$

Q.17

If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$ then the value of $\frac{\mathrm{p}}{\mathrm{p}-\mathrm{a}} + \frac{\mathrm{q}}{\mathrm{q}-\mathrm{b}} + \frac{\mathrm{r}}{\mathrm{r}-\mathrm{c}}$ is equal to

(A) -1

(B) 1

(C) -2

(D) 2

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Q.18

If $\begin{vmatrix} X^{n} & X^{n+2} & X^{n+3} \\ Y^{n} & Y^{n+2} & Y^{n+3} \\ Z^{n} & Z^{n+2} & Z^{n+3} \end{vmatrix} = (y-z)(z-x)(x-y)\left(\frac{1}{X}+\frac{1}{Y}+\frac{1}{Z}\right)$ then

(A) $n=2$

(B) $n=-2$

(C) $n=-1$

(D) $n=1$

Q.19

$[a, b, c, d>0, x \in R$ and
$\left(a^2+b^2+c^2\right)x^2-2(ab+bc+cd)x+b^2+c^2+d^2 \leq 0$
Then, $\begin{vmatrix} 33 & 14 & \log a \\ 65 & 27 & \log b \\ 97 & 40 & \log c \end{vmatrix}$ is equal to

(A) 1

(B) -1

(C) 2

(D) 0

Q.20

If $A\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$, then the value of the determinant $\left|A^{2009}-5A^{2008}\right|$ is

(A) -6

(B) -5

(C) -4

(D) 4

Answers & Solutions

Q.1 Answer:

$8\cos^3\theta-4\cos\theta+6$

Solution:

Given that, $\Delta=\begin{bmatrix} C & 1 & 0 \\ 1 & C & 1 \\ 6 & 1 & C \end{bmatrix}=C\left(C^2-1\right)-1(C-6)$

$\Rightarrow \Delta=2\cos\theta\left(4\cos^2\theta-1\right)-(2\cos\theta-6)$

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$= 8\cos^3\theta - 4\cos\theta + 6$

$(\because \mathrm{C} = 2\cos\theta$ **given** $)$

Q.2 Answer:

$-2 - \sqrt{2}$

Solution:

Expanding by Sarrus rule,

$$\begin{vmatrix} 1 & e^{i\pi/3} & e^{i\pi/4} \\ e^{-i\pi/3} & 1 & e^{i2\pi/3} \\ e^{-i\pi/4} & e^{-i2\pi/3} & 1 \end{vmatrix} = 1 + e^{i\pi/3} \times e^{i2\pi/3} \times e^{-i\pi/4} +$$

$e^{-i\pi/3} \times e^{-i2\pi/3} \times e^{-i\pi/4} - \left(e^{i2\pi/4} \times e^{-i\pi/4} + e^{-i\pi/3} \times\right.$

$= 1 + e^{i3\pi/4} + e^{-i3\pi/4} - (1 + 1 + 1)$

$\Rightarrow \Delta = -2 + e^{3ii\pi/4} + e^{-3ii\pi/4} = -2 + 2\cos(3\pi/4) = -2 - \sqrt{2} = -(2 + \sqrt{2})$

Q.3 Answer:

is greater than -8

Solution:

$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} > 0$$

$\Rightarrow a(bc - 1) - 1(c - 1) + 1(1 - b) > 0$

$\Rightarrow \quad abc - a - c + 1 + 1 - b > 0$

$\Rightarrow \quad abc + 2 - (a + b + c) > 0$

$\Rightarrow \quad abc > (a + b + c) - 2$

Let; $a = -1; b = 0 \& c = 1$

Then; $0 > -2$ **[which is correct] Hence,** $abc = 0 \Rightarrow abc > -8$

Q.4 Answer:

0

Solution:

We can write $\Delta = \Delta_1 + y_1\Delta_2$**, where** $\Delta_1 = \begin{vmatrix} 1 & 1 + x_1y_2 & 1 + x_1y_3 \\ 1 & 1 + x_2y_2 & 1 + x_2y_3 \\ 1 & 1 + x_3y_2 & 1 + x_3y_3 \end{vmatrix}$ **and**

$\Delta_2 = \begin{vmatrix} x_1 & 1 + x_1y_2 & 1 + x_1y_3 \\ x_2 & 1 + x_2y_2 & 1 + x_2y_3 \\ x_3 & 1 + x_3y_2 & 1 + x_3y_3 \end{vmatrix}$

In Δ_1**, use** $C_2 \to C_2 - C_1$ **and** $C_3 \to C_3 - C_1$

so that, $\Delta_1 = \begin{vmatrix} 1 & x_1y_2 & x_1y_3 \\ 1 & x_2y_2 & x_2y_3 \\ 1 & x_3y_2 & x_3y_3 \end{vmatrix} = 0$

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$[\because C_2$ and C_3 are proportional $]$ In Δ_2, us $C_2 \to C_2 - y_2 C_1$ and $C_3 \to C_3 - y_3 C_1$ to get

$$\Delta_2 = \begin{vmatrix} x_1 & 1 & 1 \\ x_2 & 1 & 1 \\ x_3 & 1 & 1 \end{vmatrix} = 0 \, [\because C_2 \text{ and } C_3 \text{ are identical}]$$

$\therefore \Delta = 0$

Q.5 Answer:

2

Solution:

$$f(x) = \begin{vmatrix} 1+\left(a^2+b^2+c^2+2\right)x & \left(1+b^2\right)x & \left(1+c^2\right)x \\ 1+\left(a^2+b^2+c^2+2\right)x & 1+b^2x & \left(1+c^2\right)x \\ 1+\left(a^2+b^2+c^2+2\right)x & \left(1+b^2\right)x & 1+c^2x \end{vmatrix}$$

Applying, $C_1 \to C_1 + C_2 + C_3$

$$= \begin{vmatrix} 1 & \left(1+b^2\right)x & \left(1+c^2\right)x \\ 1 & 1+b^2x & \left(1+c^2\right)x \\ 1 & \left(1+b^2\right)x & 1+c^2x \end{vmatrix}$$

$\therefore a^2+b^2+c^2+2 \neq 0$

$$f(x) = \begin{vmatrix} 0 & x-1 & 0 \\ 0 & 1-x & x-1 \\ 1 & \left(1+b^2\right)x & 1+c^2x \end{vmatrix}$$

Applying $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$

$f(x) = (x-1)^2$ **Hence degree** $= 2$.

Q.6 Answer:

21

Solution:

$$\begin{bmatrix} x^2+x & 3x-1 & -x+3 \\ 2x+1 & 2+x^2 & x^3-3 \\ x-3 & x^2+4 & 3x \end{bmatrix} = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_7 x^7$$

Put $x = 0 \Rightarrow \begin{vmatrix} 0 & -1 & 3 \\ 1 & 2 & -3 \\ -3 & 4 & 0 \end{vmatrix} = a_0 \quad \Rightarrow \quad a_0 = 21$

Q.7 Answer:

Negative

Solution:

$R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$, (reduces the

0.8

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$$\begin{vmatrix} x^2-2x+1 & x-1 & 0 \\ 2x-2 & x-1 & 0 \\ 3 & 3 & 1 \end{vmatrix}$$

0

$= (x-1)^3 - 2(x-1)^2 = (x-1)^2(x-1-2)$

$= (\mathrm{x}-1)^2(\mathrm{x}-3)$, **which is clearlynegative for** $\mathrm{x} < 1$

Q.8 Answer:

[z]

Solution:

Since, $-1 < x < 0 \quad \therefore \quad [x] = -1$

$0 < y < 1 \quad \therefore [y] = 0, \quad 1 < z < 2 \quad \therefore [z] = 1$

$\therefore$ **Given determinant** $= \begin{vmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ -1 & 0 & 2 \end{vmatrix} = 1 = [\mathrm{z}]$

Q.9 Answer:

$-\frac{1}{2}$

Solution:

We know that $\det(\mathrm{A}) = \frac{1}{\det(\mathrm{A}^{-1})}$

$\mathrm{A}^{-1} = \begin{bmatrix} 1 & -2 \\ -2 & 2 \end{bmatrix}$

$\Rightarrow |\mathrm{A}^{-1}| = \begin{vmatrix} 1 & -2 \\ -2 & 2 \end{vmatrix} = 2 - 4 = -2$

$\Rightarrow \det(\mathrm{A}) = -\frac{1}{2}$

Q.10 Answer:

$(14)^4$

Solution:

We know that $\mathrm{adj}(\mathrm{adj}\ \mathrm{A}) = |\mathrm{A}|^{\mathrm{n}-2}\ \mathrm{A}$, **if** $|\mathrm{A}| \neq 0$, **provided order of** A **is** n

$\therefore \mathrm{adj}(\mathrm{adj}\ A) = |A|A (\mathrm{as}\, n = 3)$

$\therefore \det(\mathrm{adj}(\mathrm{adj}\ A)) = |A|^3 \det A = |A|^4$

But $|\mathrm{A}| = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix} = 14$

$\therefore \det(\mathrm{adj}(\mathrm{adj}\ A)) = (14)^4$

Q.11 Answer:

$-f(\alpha)f(\beta)f(\gamma)$

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Solution:

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -\left(a^3+b^3+c^3-3abc\right)$$
$$= -(a+b+c)\left(a+b\omega^2+c\omega\right)\left(a+b\omega+c\omega^2\right)$$
$$= -f(\alpha)f(\beta)f(\lambda)$$
$$\left[\because \alpha = 1, \beta = \omega, = \omega^2\right]$$

Q.12 Answer:

2

Solution:

Given matrix is :

$$\begin{vmatrix} x^2 & -2x & -2\omega^2 \\ 2 & \omega & -\omega \\ 0 & \omega & 1 \end{vmatrix} = 0$$

By $C_2 \to C_2 + C_3$, **we get**

$$\Rightarrow \begin{vmatrix} x^2 & -2x-2\omega^2 & -2\omega^2 \\ 2 & 0 & -\omega \\ 0 & 1+\omega & 1 \end{vmatrix} = 0$$
$$\Rightarrow \begin{vmatrix} x^2 & -2x-2\omega^2 & -2\omega^2 \\ 2 & 0 & -\omega \\ 0 & -\omega^2 & 1 \end{vmatrix} = 0$$
$$\left[\because \quad 1+\omega = -\omega^2\right]$$
$$\Rightarrow \omega^2 \begin{vmatrix} x^2 & -2\omega^2 \\ 2 & -\omega \end{vmatrix} + 1 \begin{vmatrix} x^2 & -2x-2\omega^2 \\ 2 & -0 \end{vmatrix} = 0$$
$$\Rightarrow \omega^2\left(-\omega x^2+4\omega^2\right) - \left(-4x-4\omega^2\right) = 0$$
$$\Rightarrow -x^2+4\omega+4x+4\omega^2 = 0$$
$$\Rightarrow -x^2+4x-4 = 0$$
$$\Rightarrow x^2-4x+4 = 0$$
$$\Rightarrow (x-2)^2 = 0$$
$$\Rightarrow x = 2$$

Q.13 Answer:

0

Solution:

$$\text{Let } \Delta = \begin{vmatrix} \cos^2 54^\circ & \cos^2 36^\circ & \cot 135^\circ \\ \sin^2 53^\circ & \cot 135^\circ & \sin^2 37^\circ \\ \cot 135^\circ & \cos^2 25^\circ & \cos^2 65^\circ \end{vmatrix}$$
$$= \begin{vmatrix} \cos^2 54^\circ & \sin^2 54^\circ & -1 \\ \cos^2 37^\circ & -1 & \sin^2 37^\circ \\ -1 & \cos^2 25^\circ & \sin^2 25^\circ \end{vmatrix}$$

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$C_1 \to C_1 + C_2 + C_3$

$$= \begin{vmatrix} \cos^2 54^\circ + \sin^2 54^\circ - 1 & \sin^2 54^\circ & -1 \\ \cos^2 37^\circ - 1 + \sin^2 37^\circ & -1 & \sin^2 37^\circ \\ -1 + \cos^2 25^\circ + \sin^2 25^\circ & \cos^2 25^\circ & \sin^2 25^\circ \end{vmatrix}$$

$$= \begin{vmatrix} 0 & \sin^2 54^\circ - 1 \\ 0 & -1 \sin^2 37^\circ \\ 0 & \cos^2 25^\circ \sin^2 25^\circ \end{vmatrix} = 0$$

Q.14 Answer:

$\mathrm{n} + 1$

Solution:

Taking x^5 common from last row, we get, $x^5 \begin{vmatrix} x^n & x^{n+2} & x^{2n} \\ 1 & x^a & a \\ x^n & x^{a+1} & x^{2n} \end{vmatrix} = 0 \forall x \in R$

$\Rightarrow a + 1 = n + 2 \Rightarrow a = n + 1$

(as it will make first and third row identical)

Q.15 Answer:

$\sin x = 0$

Solution:

Given determinant

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(n-1)x & \cos nx & \cos(n+1)x \\ \sin(n-1)x & \sin nx & \sin(n+1)x \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 + a^2 - 2a\cos x & a & a^2 \\ 0 & \cos nx & \cos(n+1)x \\ 0 & \sin nx & \sin(n+1)x \end{vmatrix}$$

$= 0$

By applying $C_1 \to C_1 + C_3 - 2\cos x C_2$

By expanding $1 + a^2 - 2a\cos x[\cos nx \sin(n+1)x$

$-\sin nx \cos(n+1)x] = 0$

Now, $1 + a^2 - 2a\cos x \sin(n+1-n)x = 0$

$\Rightarrow 1 + a^2 - 2a\cos x \sin x = 0$

$\sin x = 0$ **or** $\cos x = \frac{1+a^2}{2a}$

As $a \neq 1 \therefore \frac{1+a^2}{2a} > 1$

$\Rightarrow \cos x > 1$ **It is not possible.**

$\therefore \sin x = 0$

Q.16 Answer:

$n(B) = n(C)$

Solution:

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C cannot be the empty set becanse, for instance

$-1 = \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} \in C$. We also have $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2$

So $A \neq B \cup C$. In general, the determinant

With the a's being 0 or 1, equal 1 only if $a_{11}a_{22}^{a}gg = 1$ and the remaining tems are zero; if $a_{12}a_{2a}a_{21} = 1$ and the remaining tems are zero; or if $a_{19}a_{21}a_{92} = 1$ and the remaining tems are zero. Since there are three similar relations for determinants that equal -1. We must have $n(b) = n(c)$

Q.17 Answer:

2

Solution:

$$\begin{vmatrix} \mathrm{p} & \mathrm{b} & \mathrm{c} \\ \mathrm{a} & \mathrm{q} & \mathrm{c} \\ \mathrm{a} & \mathrm{b} & \mathrm{r} \end{vmatrix} = 0$$

$\mathrm{R}_1 \to \mathrm{R}_1 - \mathrm{R}_2, \mathrm{R}_2 \to \mathrm{R}_2 - \mathrm{R}_3$

$$\begin{vmatrix} \mathrm{p}-\mathrm{a} & \mathrm{b}-\mathrm{q} & 0 \\ 0 & \mathrm{q}-\mathrm{b} & \mathrm{c}-\mathrm{r} \\ \mathrm{a} & \mathrm{b} & \mathrm{r} \end{vmatrix} = 0$$

$(\mathrm{p}-\mathrm{a})[\mathrm{r}(\mathrm{q}-\mathrm{b}) - \mathrm{b}(\mathrm{c}-\mathrm{r})] - (\mathrm{b}-\mathrm{q})[-\mathrm{a}(\mathrm{c}-\mathrm{r})] = 0$

$\mathrm{r}(\mathrm{p}-\mathrm{a})(\mathrm{q}-\mathrm{b}) + \mathrm{b}(\mathrm{p}-\mathrm{a})(\mathrm{r}-\mathrm{c}) + \mathrm{a}(\mathrm{q}-\mathrm{b})(\mathrm{r}-\mathrm{c}) = 0$

$\Rightarrow \frac{r}{r-c} + \frac{b}{q-b} + \frac{a}{p-a} = 0$

$\Rightarrow \frac{r}{r-c} = -\frac{b}{q-b} - \frac{a}{p-a}$

Now, consider $\frac{\mathrm{p}}{\mathrm{p}-\mathrm{a}} + \frac{\mathrm{q}}{\mathrm{q}-\mathrm{b}} + \frac{\mathrm{r}}{\mathrm{r}-\mathrm{c}}$

$= \frac{p}{p-a} + \frac{q}{q-b} - \frac{b}{q-b} - \frac{a}{p-a}$

$= 1 + 1 = 2$

Q.18 Answer:

$n = -1$

Solution:

Order of the variables will be a constant. Order of the determinant $= n + n + 2 + n + 3 = 3n + 5$

Order of the answer $= 3 - 1 = 2$ Hence $3n + 5 = 2$ and therefore $n = -1$

Q.19 Answer:

0

Solution:

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Given,

$$\left(a^2+b^2+c^2\right)x^2-2(ab+bc+cd)x+b^2+c^2$$

$$\Rightarrow \quad (ax-b)^2+(bx-c)^2+(cx-d)^2\le 0$$

$$\Rightarrow \quad (ax-b)^2+(bx-c)^2+(cx-d)^2=0$$

$$\Rightarrow \frac{b}{a}=\frac{c}{b}=\frac{d}{c}=x$$

$$\Rightarrow b^2=ac \text{ or } 2\log b=\log a+\log c$$

Now, $\Delta=\begin{vmatrix} 33 & 14 & \log a \\ 65 & 27 & \log b \\ 97 & 40 & \log c \end{vmatrix}$

Apply $R_1\to R_1+R_3$

$$\Delta=\begin{vmatrix} 130 & 54 & \log a+\log c \\ 65 & 27 & \log b \\ 97 & 40 & \log c \end{vmatrix}=0$$

Now, $R_1\to R_1-2R_2$

$$\Delta=\begin{vmatrix} 0 & 0 & 0 \\ 65 & 27 & \log b \\ 97 & 40 & \log c \end{vmatrix}=0$$

Q.20 Answer:

-6

Solution:

Given, $A=\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

$$\Rightarrow \quad |\mathrm{A}|=5-6=-1$$

$$\therefore \quad \left|\mathrm{A}^{2009}-5\,\mathrm{A}^{2008}\right|=\left|\mathrm{A}^{2008}\right|\,|\mathrm{A}-5\mathrm{I}|$$

$$=(-1)^{2008}\left|\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}-\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}\right|=\begin{vmatrix} -4 & 2 \\ 3 & 0 \end{vmatrix}=-6$$

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